# Modelling

Higher · Design · Worksheet

D&M WITH MR KELLY

Name _____ Class _____ Date _____

**By the end of this worksheet I can…**

- ☐ explain how models are used to generate and explore, test and refine, and communicate
- ☐ describe the purpose of, and information gained from, test rigs and prototypes
- ☐ explain the use of computer models and simulations
- ☐ describe the benefits and drawbacks of rapid prototyping

## I DO — Worked example

Read how a full-mark answer is built

**1 Describe how physical models can be used during the design of a product.** (4 marks)

**Full-mark answer** each ✓ is one mark

Sketch models made quickly from card let the designer try out several forms and check proportions early on ✓. A block model can be held to test the grip and button positions against hand sizes ✓. A test model or rig can check strength and stability, so faults are found before tooling is made ✓. A scale model shows the size of a large product relative to a person ✓.

**Examiner tip:** 2025 commentaries credited four distinct purposes; naming types alone and CAD models scored 0.

**Why marks are not awarded**

*"Sketch models, block models, prototypes and CAD models."*

0 marks: naming model types scores nothing, and computer models are not physical models.

## WE DO — Guided practice

Use the starters and word bank to help you

**2 Discuss how designers use a variety of models during the design process. Link each to the stage, purpose and information gained.** (8 marks)

**Sentence starters**

Scale models: …

Computer models and simulations: …

**Word bank**

models · ergonomics · explore · proportion · client · sketch · generate · refine

Tip: Marked in bands: cover several stages, each with an example and why it is used.

**YOU DO** **On your own** No notes. Check your answers afterwards.

### 3 Quick check: tick one box for each.

**a)** An architect shows a new school building to the council. Which model?
☐ Scale model ☐ Sketch model ☐ Block model ☐ Computer-generated model

**b)** Which is a benefit of simulation?
☐ It uses the real material, so results are exact ☐ It needs no computer or special software ☐ Testing without risk to users or the environment ☐ It is always cheaper and quicker than sketching

**c)** Why are sketch models used early in the design process?
☐ They show exactly what the final finish will be ☐ They are quick and cheap to make and change ☐ They are fully working and can be user tested ☐ They are used to set up mass production

**d)** Which model would a designer use to test the comfort of a handle grip?
☐ Scale model ☐ Exploded drawing ☐ Sketch model of a different product ☐ Block model

**4 Key terms: write the letter of the matching description next to each term.**

| | | |
|---|---|---|
| 1. **Sketch model** | ____ | **A** Solid, non-working model showing shape and form; often used to test ergonomics. |
| 2. **Block model** | ____ | **B** A model that looks and/or works like the final product, used for final testing before production. |
| 3. **Scale model** | ____ | **C** Easy to change, realistic, can produce drawings and link to CAM/3D printing. |
| 4. **Computer-generated model benefit** | ____ | **D** Model at a different scale for objects too large or small to model full size. |
| 5. **Prototype** | ____ | **E** Quick, cheap model from scrap materials to generate and explore ideas. |

**5 Explain what is meant by 'Simulation'.** (2 marks)

**6 Describe one drawback of rapid prototyping.** (2 marks)

**Check and reflect**
Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each "I can" statement above. Which one still needs work?